

Renormalized Branch Amplitudes for the Square-Billiard Abel Trace

Route-A structural certificate C162: first improvement

Abstract

For the unit-square Dirichlet Abel half-wave trace, the first improvement promotes the local $\epsilon^{3/2}$ branch calculation to a positive-time full-trace theorem. A finite-shell split and a uniform two-dimensional $|m|^{-3}$ majorant justify passage through the infinite dual sum. The resulting coefficient is the principal-branch constant times aggregate source shell multiplicity. Negative-time conjugation and the systematic coincident-pole statement are deferred to the final round.

Keywords: square billiard; Abel half-wave trace; full-trace limit; uniform domination; shell multiplicity.

中文摘要：第一轮改进把单项分支计算提升为正时间的全迹定理。有限壳层分离与二维 $|m|^{-3}$ 一致可求和控制允许穿过无限对偶和，极限系数精确记录源壳层总重数。负时间共轭和重合极点的系统结论留到最终轮次。

关键词：正方形台球；阿贝尔半波迹；全迹极限；一致控制；壳层重数。

1 Positive-time full-trace theorem

At $t_N = 2\sqrt{N}$, let $r_2^{\text{src}}(N) = \#\{m \in \mathbb{Z}^2 : |m|^2 = N\}$. Then

$$\lim_{\epsilon \downarrow 0} \epsilon^{3/2} W_D(\epsilon - it_N) = \frac{e^{i\pi/4} r_2^{\text{src}}(N)}{8\pi N^{1/4}}.$$

Each matching vector gives the displayed principal-branch constant. For the remainder, fix t_N and $0 < \epsilon \leq 1$. Outside a radius with $4|m|^2 \geq 2(t_N^2 + 1)$,

$$|s^2 + 4|m|^2| \geq 2|m|^2.$$

Thus the tail is uniformly dominated by a constant times $\sum_{m \neq 0} |m|^{-3} < \infty$. The finitely many nonmatching shells stay bounded, so dominated convergence proves the full-trace limit.

2 First-round boundary

The coefficient aggregates clean source families, not isolated stability weights. This round does not yet claim the negative-time or coincident-pole receipts. No target trace/divisor law, arithmetic Euler factor, root number, Hilbert–Pólya construction, or Route B is claimed.

Declarations

All claim-bearing data and code are packaged; no external dataset, subjects, or funding is involved. No competing interest is known. An AI coding assistant supported drafting and code review but was not an external reviewer.